TrafficSense — Multi-Agent Smart City Traffic Management System

Capstone Project | School of Computer Science & Engineering | VIT University

Student Information

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Project Title	TrafficSense: A Multi-Agent Smart City Traffic Management System
Role	Computer Vision & ML Lead — responsible for YOLOv8 model training, custom Indian dataset creation, augmentation pipeline, and real-time inference integration
Academic Year	FALL SEM 2026-27
Faculty Mentor	Dr. Mohd Irfan, Assistant Professor Sr. Grade-1, SCOPE, VIT-AP University

Contribution Overview

My primary responsibility was the entire Computer Vision and Machine Learning pipeline. This includes selecting the detection model, building the custom Indian traffic dataset, designing and executing all training runs, integrating YOLO-World for zero-shot detection, and deploying the final model into the live inference pipeline. The model I trained achieves a final mAP50 of **70.5%** with auto-rickshaw detection at **99.5% accuracy** — a class that standard COCO models cannot detect at all.

Detailed Technical Contributions

1. Model Selection & Architecture

- Evaluated YOLOv8n, YOLOv8s, and YOLOv8m variants for accuracy vs. VRAM trade-off on the RTX 4050 6GB GPU.
- Selected YOLOv8m (25M parameters) as the final model — provides significantly better accuracy than YOLOv8s while fitting within 6GB VRAM at 640px.

• Integrated YOLO-World (open-vocabulary zero-shot model) as an alternate detection engine allowing text-based class specification.

2. Custom Indian Traffic Dataset Creation

• Defined a comprehensive 24-class taxonomy specific to Indian roads: car, motorcycle, bus, truck, autorickshaw, bicycle, ambulance, police_vehicle, tractor, pedestrian, traffic_light, traffic_sign, scooter, e-rickshaw, van, tempo, animal, pushcart, water_tanker, excavator, crane, jeep, mini_bus, fire_engine.

• Sourced and labeled images from multiple Indian traffic datasets and integrated them into a unified YOLO-format dataset.

• Used Roboflow API to download and merge 4 additional Indian traffic datasets, growing the corpus to 42,652 training images after augmentation.

• Wrote the class remapping pipeline (src/perception/download_datasets.py) to normalise heterogeneous class names from different sources into our 24-class scheme.

3. Data Augmentation Pipeline

• Built a heavy augmentation pipeline (src/perception/augment_dataset.py) using the Albumentations library.

• Applied 10+ augmentation strategies: random fog, simulated rain, ISO noise, motion blur, perspective warp, brightness/contrast variation, and HSV shifts — simulating Indian monsoon and night driving conditions.

• Multiplied the base dataset ×15, converting ~2,840 raw images into 42,652 synthetic training variants.

4. Model Training — Nuclear Run + Fine-Tuning

• Executed a large-scale nuclear training run on 42,652 images at 832px resolution (YOLOv8m, AdamW optimizer, 9 full epochs = 380,000+ image passes).

• Designed and executed a targeted fine-tuning stage (src/perception/fast_finetune.py) from the pre-trained checkpoint on a curated 1,500-image balanced subset at 640px.

• Achieved early stopping at epoch 57 (patience=30), with best weights saved at epoch 27.

• Final model performance: **mAP50 = 70.5%, mAP50-95 = 67.4%**, auto-rickshaw mAP50 = 99.5%.

• Auto-deployed best.pt to models/yolo/best.pt for direct use by the dashboard.

5. Inference Pipeline

• Built the real-time inference pipeline (src/perception/inference_pipeline.py) supporting both file-based and live webcam inputs.

• Integrated ByteTrack multi-object tracking to assign persistent vehicle IDs across frames.

• Built the CongestionAnalyzer module to convert per-frame detections into congestion scores.

■ Training Results Summary

Class	mAP50	Precision	Recall

All Classes (Average)	70.5%	76.2%	61.4%
Car	78.7%	86.1%	75.0%
Auto-rickshaw	99.5%	62.8%	100%
Bus	99.5%	50.1%	100%
Pedestrian	99.5%	90.9%	100%
Police Vehicle	99.5%	61.8%	100%
Van	99.5%	100%	—

■ Deliverables & Files

File / Module	Description	Status
src/perception/train_yolo.py	Primary YOLOv8 training script	✔ Complete
src/perception/fast_finetune.py	Targeted fine-tuning from checkpoint	✔ Complete
src/perception/augment_dataset.py	Albumentations augmentation pipeline	✔ Complete
src/perception/download_datasets.py	Roboflow multi-dataset downloader	✔ Complete
src/perception/full_pipeline.py	End-to-end data → train pipeline	✔ Complete
src/perception/inference_pipeline.py	Real-time YOLOv8 inference + tracking	✔ Complete
src/perception/tracker.py	ByteTrack vehicle tracker	✔ Complete
src/perception/congestion_analyzer.py	Congestion level computation	✔ Complete
models/yolo/best.pt	Trained model weights (52MB, YOLOv8m)	✔ Deployed
data/processed/unified_indian/	Custom 24-class Indian traffic dataset	✔ Complete

■ Technologies & Tools Used

Framework	Ultralytics YOLOv8 (v8.4.129)
Model	YOLOv8m — 25.8M parameters, 79.1 GFLOPs
Zero-Shot Model	YOLO-World with CLIP embeddings

Augmentation	Albumentations v1.x
Dataset Platform	Roboflow Universe API
Hardware	NVIDIA GeForce RTX 4050 Laptop (6GB VRAM), CUDA 12.1
Language	Python 3.12, PyTorch 2.5.1
Training Time	~18 hours total (nuclear run + fine-tune)

■ Declaration

I, C Bhargav Kumar (23BCE8195), hereby declare that the contributions described in this report — including model selection, dataset construction, augmentation pipeline, all training runs, and inference integration — are my individual work carried out as part of the TrafficSense Capstone Project.

Signature: _____ Date: _____